

Often there was a need for skilled workers, such as masons, carpenters, or blacksmiths, and if there was no skill available, engineers ensured that they taught labourers what was required. It was just a matter of time before knowledge was transferred between workers who would often travel between sites as and when their expertise was required, wrote Kerr.

Cultural differences

British engineers and contractors often faced resistance when they tried to change the technologies and practices that were already in use. Henry Fowler, contracted to build the first part of the Great Indian Peninsula Railway (GIPR) line, was recorded complaining of labourers who refused to begin work at 6am, instead preferring the usual 8am. Workers also often refused to take on jobs alongside those of a different caste.

Then there is the story as told by engineer Thomas Going, which, wrote Kerr, “is so evocative as to suggest some embellishment”: “It has been often attempted to introduce the wheelbarrow mode of work, but with little success. The basket of antiquity—probably antediluvian – still holds its own. I have heard of an enthusiast in wheelbarrows who, having exhausted his morning energy in the fond endeavour to restrain a gang of coolies from using the objectionable basket, had the mortification, on making his evening tour of inspection, to find them carrying the wheelbarrows on their heads, in the belief that it was only a convenient modification of the principle.”

Building bridges

One of the greatest challenges for the early engineers and the workers were bridging the rivers, from the Indus and Ganges, to the Narmada, Jamuna, Sutlej, Beas, Godavari, and Hooghly. While Britain had seen significant development in bridge-building for the railways, the technology seemed nearly inapplicable in India.

These almost-gentle rivers with low water levels would become raging torrents during the monsoon, as would the Himalayan-fed rivers, when the ice melted. They would erode rock and weaken bridge